Predicting the Probability and Cost of Accidents in Austin

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Agenda

1. Relevance of Problem and Solution Urgency
2. Data Description
3. Exploratory Model: Tweedie XGBoost
4. LightGBM
5. Hurdle Model
6. Takeaways

What’s the problem?

Motor-vehicle crashes cost the local government approximately 35.24 million USD in 2022 dollars

One reportable crash occurred every 57 seconds

Identifying when and where crashes are most likely to occur can therefore help design targeted mitigation strategies.

What’s our solution?

Build **a spatiotemporal risk surface** that quantifies the **probability and expected cost** of accidents across Austin

Applications:

- Insurance premiums
- Rideshare platforms **compensation** systems

The Data

The Austin Crash Report Dataset

City of Austin's Open Data Portal

47 features 220,000 records

Updated daily

Temporal Variables

Hour, day of week, month

Spatial Features

Lat, long

Contextual variables for road types and outcomes

Cost, Highways, people involved, severity

Engineered Features

Time features: Holiday/weekend indicators.

Spatial features: H3 grid of resolution index 8- and 1-hour bins

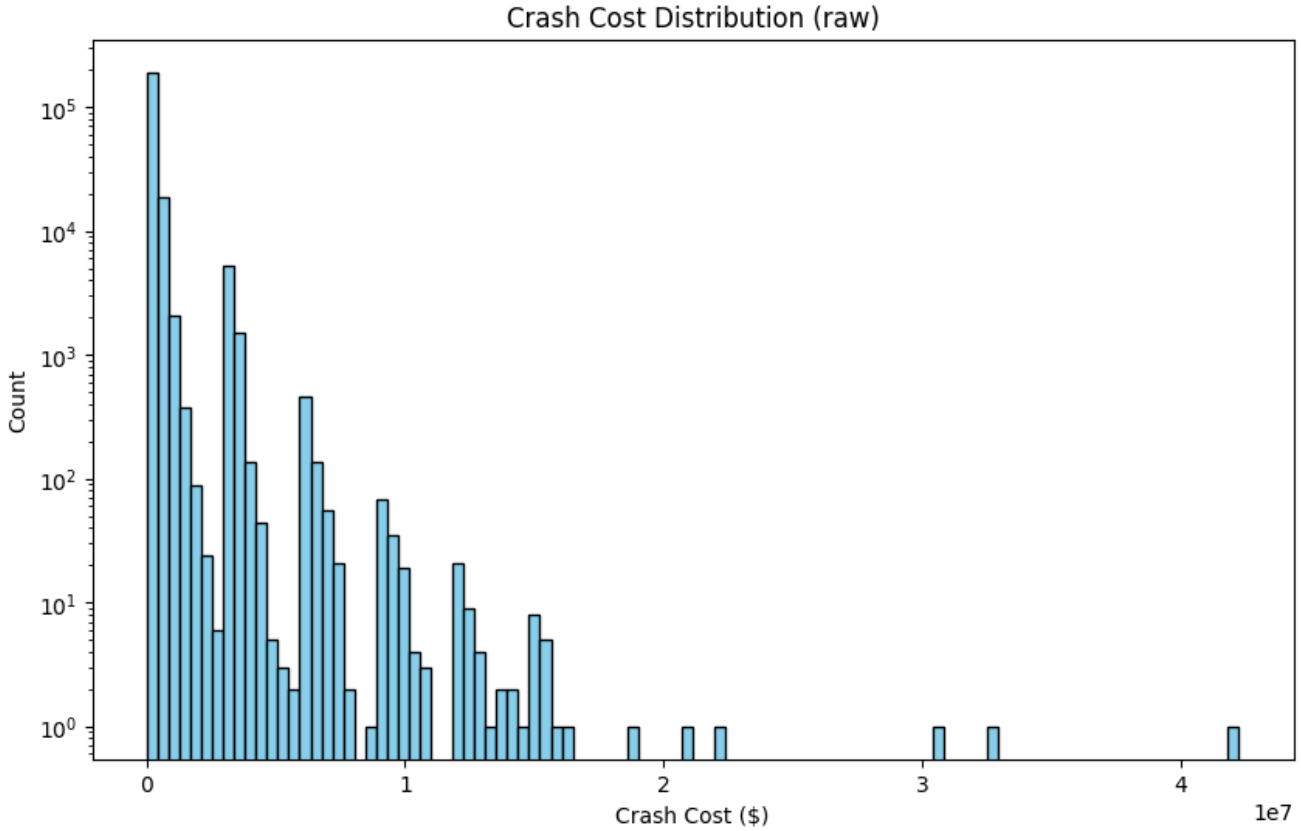
Count Features: Lagged counts, rolling means and recent-incident indicators

Weather: Rain in mm and mean temp in Fahrenheit

Negative Observations: Merge a full hex-time grid for every cell and time bin to represent where *no* crash occurred

Tweedie XGBoost

Brief Description of Model



- **Tweedie XGBoost** chosen to handle, **non-negative, highly skewed crash cost data**
- Range varies from **$20k to nearly $40M**, generating a **fat-tailed distribution**
- Model inputs are **pre-crash environmental features**, not post-crash injury or vehicle data
- Therefore, the model is used for **exploratory analysis and risk patterns**, rather than exact dollar prediction

Model Specification

Goal: Explore how pre-crash conditions relate to eventual crash cost

Predictors: Weather conditions, time, weekend/holiday, location, speed limits

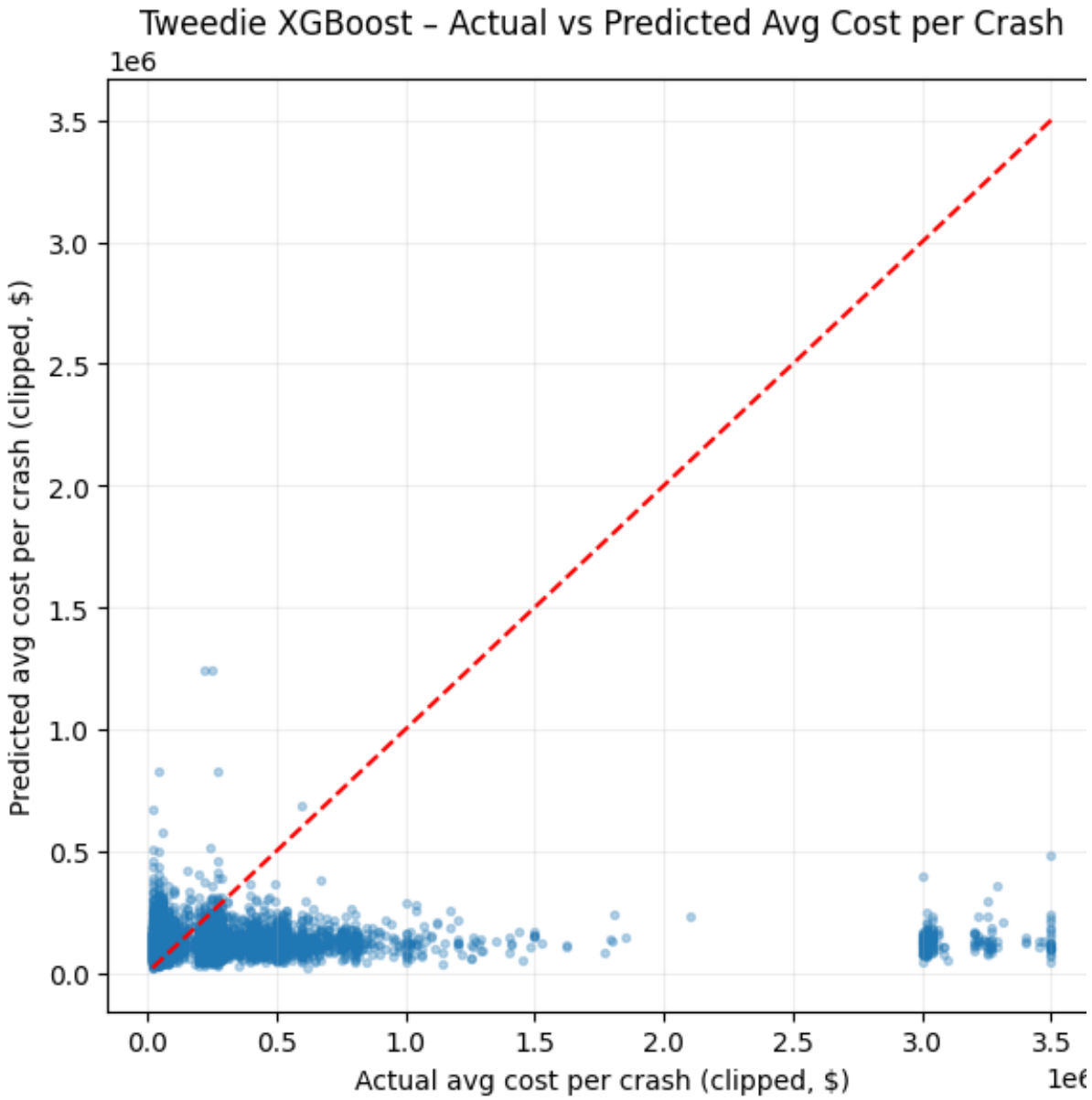
Splits: Training Split (70%)/ Validation Split (15%)/ Testing Split (15%)

Cost Stabilization: Log-transform + clipping used to stabilize extreme outliers

Important Note: Model does **not** use post-crash information (injuries, impact speed, vehicle types)

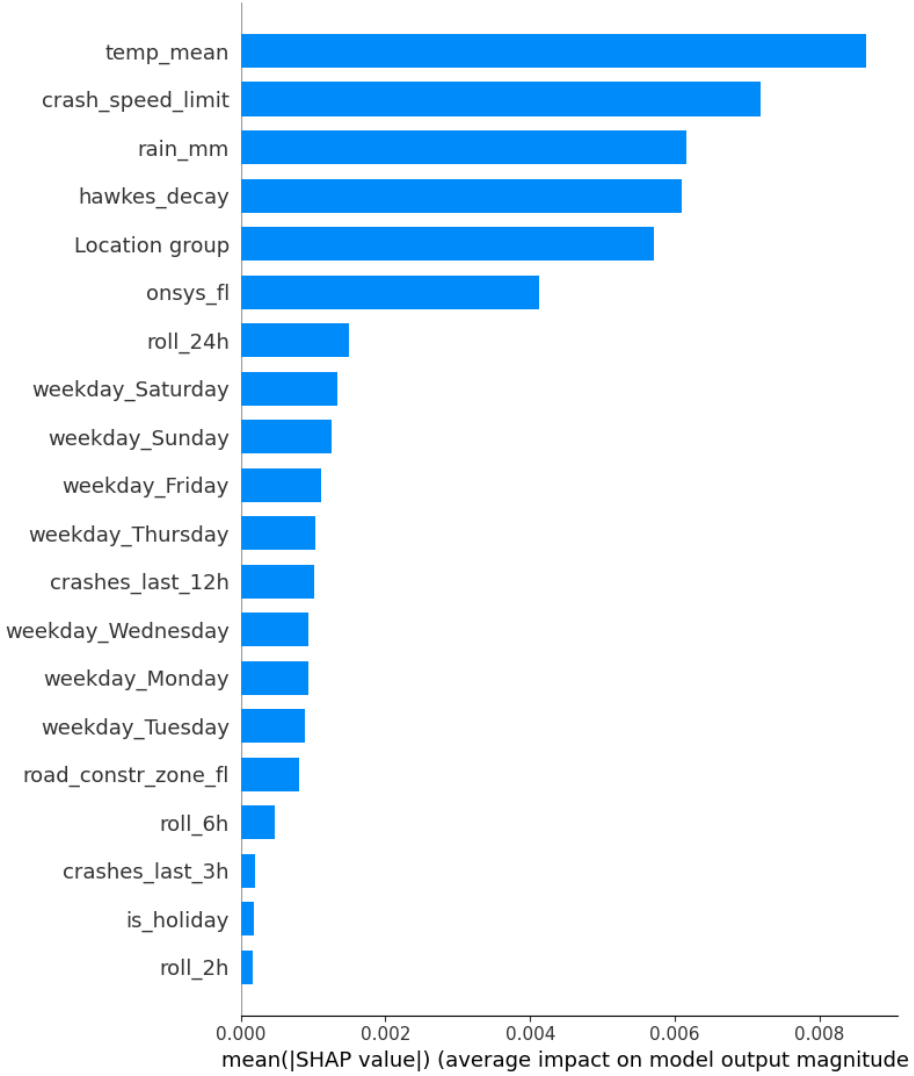
Results

- The model's **RMSE ≈ $600k** due to the **extremely heavy-tailed cost distribution**
- The model's **MAE ≈ $240k**
- The model cannot predict multi-million-dollar extreme cases
- Academic literature shows crash severity is **weakly predictable** from environmental features alone, because true severity depends on **post-crash factors** (injuries, vehicle type, impact angle, speed, EMS response), which are not included in this model



Model Explainability (SHAP Summary)

- SHAP assigns each feature a **share of influence** in the prediction, showing direction and magnitude of impact
- Reveals which **pre-crash conditions** are associated with higher crash cost
- **Higher temperatures** are associated with increased crash cost
- **Higher speed limits** and **freeway locations** push cost predictions upwards
- **Rainfall** increases expected severity
- Confirms the model is valuable for **exploratory insight**, not exact cost prediction



Binary LightGBM

Brief Description of the Algorithm

1. Prediction Task:

Binary classification problem: $\hat{p} = P(\text{crash} = 1 | \text{ engineered features})$

2. Light GBM trains an ensemble of decision trees, each one correcting the mistakes of the previous one using the gradient of the loss function.

Binary loss function: $\mathcal{L} = -[y \log(\hat{p}) + (1 - y) \log(1 - \hat{p})]$

The gradient tells the model how to correct it's mistakes: $g = \frac{d\mathcal{L}}{df}$

The Hessian (second gradient) tells LightGBM how big of a step it can take, how to compute optimal leaf values in the tree

LightGBM fits a new decision tree to reduce this gradient. This process continues until early stopping detects no further improvement on validation data.

3. Ranking Future Hex-Timebins :

We use a Sigmoid function to compute the predicted risk score for every future hex-timebin

Then we sort them from highest $\hat{p}$ to lowest and we flagged the top 5%

Model Specification

Goal:

- Predict whether a crash will occur

Predictors:

- 32 predictors, including rain, temperature, hour of day, month, a rush-hour flag, day-of-week indicators, weekend flag, location group, and speed limit

Time-based Splits:

- Train(70%)/Val(15%)/Test Split(15%) to prevent leakage and keep causality

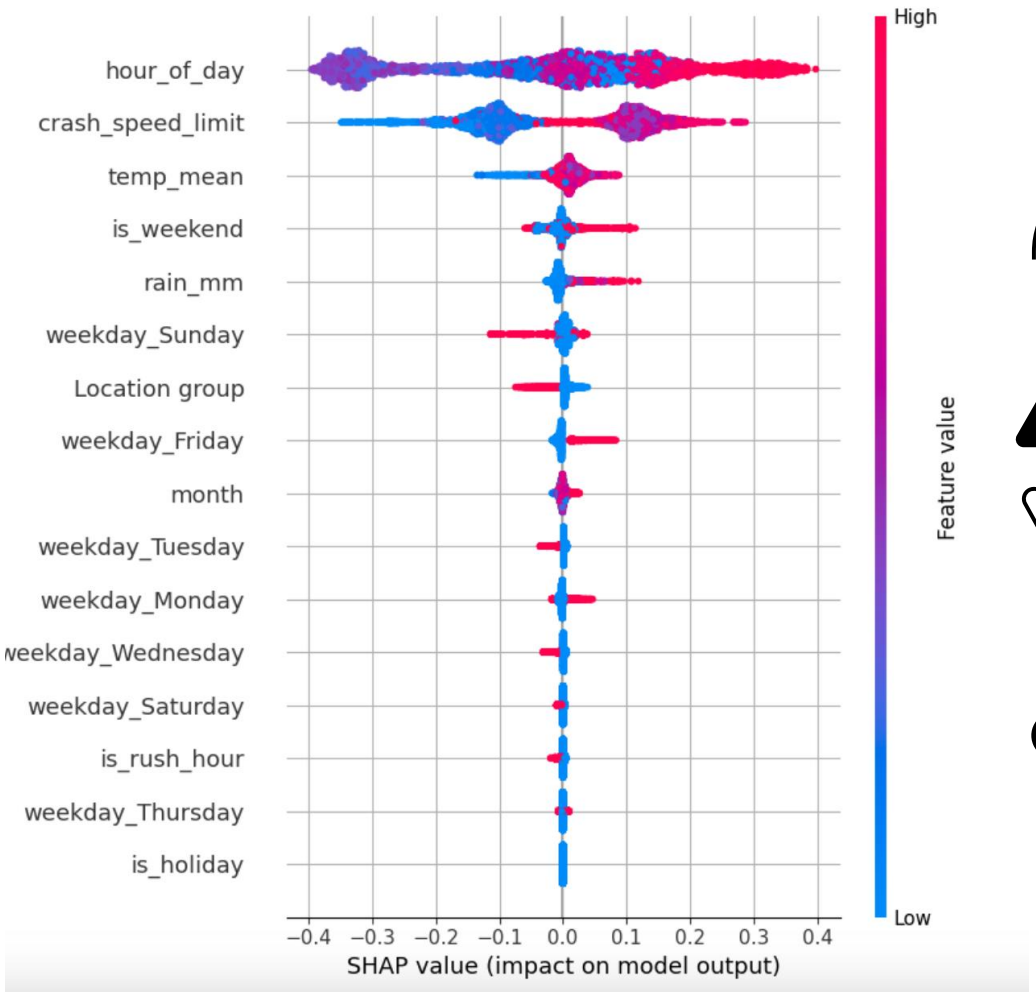
Negative sampling:

- Down sampling negatives in training (5:1); controlled sampling in evaluation (20:1)

Hyperparameters:

- Small learning rate, 31 leaves, row wise boosting, 0.8 feature and bagging fractions

Results



Crash risk peaks during busy daylight/evening hours and drops overnight

Areas with higher posted speeds contribute meaningfully to crash likelihood

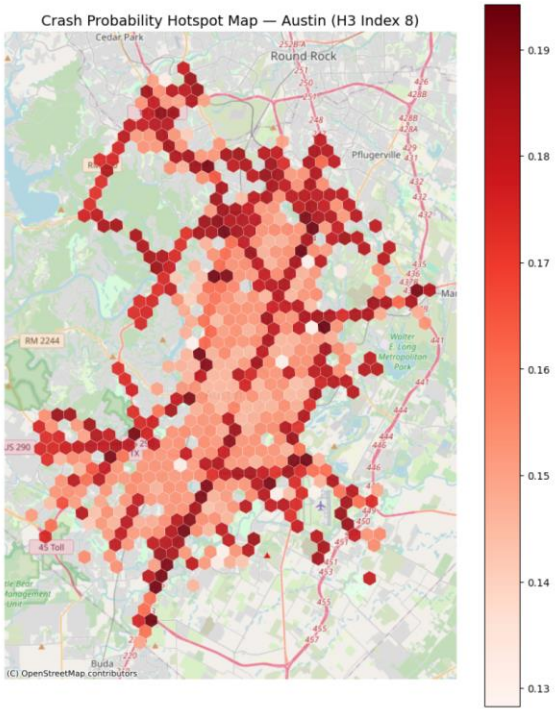
Warmer temperatures slightly increased risk. Crash patterns correlate with seasonality and daytime temperature cycles

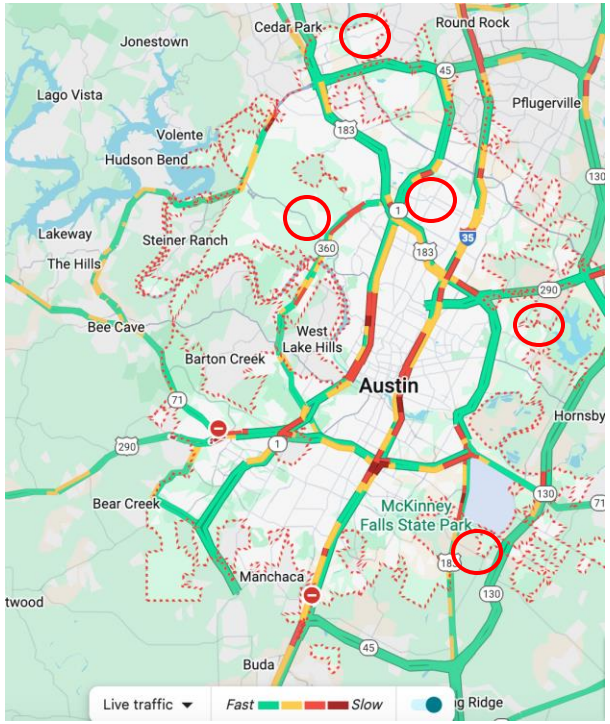
Weekend conditions moderately elevate crash probability.

Rainfall is a strong, intuitive driver of crash likelihood

Crash Probability Hotspot Map

This map shows the model's predicted crash risk over the test period.
The hotspots you see are where the model believes crashes are most likely to occur next, not just where they happened before.





- High-risk regions overlap with historically dangerous corridors like I-35 and central Austin.
- The model also elevates risk in locations where the conditions resemble high-risk patterns even if historical crash counts were low.

Performance Metrics

On top 5% bins (>22.81% risk)	
Eval Test set Metrics	**Value**
ROC AUC	64.69%
PR AUC	7.35%
Precision	8.47%
Recall	8.91%

In rare-event modeling, **raw precision/recall numbers look small**, but **the lift over baseline** is what demonstrates value. Our model lifts the crash concentration

"AUC values in the 0.6–0.7 range are common and **acceptable."** -Yu & Abdel-Aty (2014)

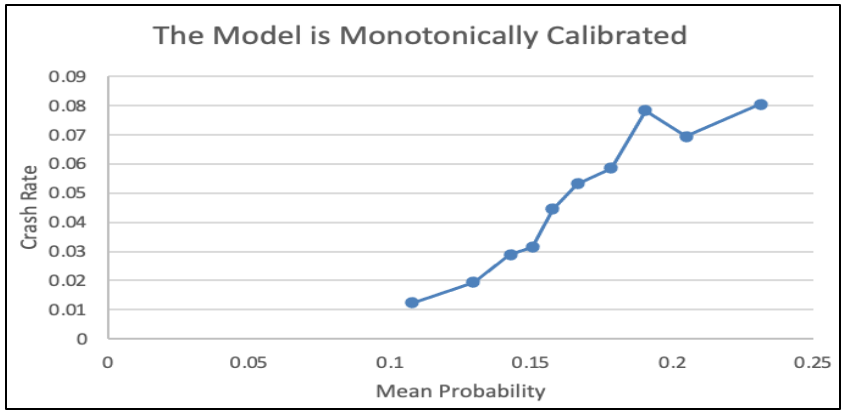
1.53x relative to random 4.8%

1.76x improvement (baseline 4.8% prevalence)

1.78x improvement (baseline 5% by design)

Full Test set Metrics	Value
ROC AUC	64.76%
PR AUC	0.3%

Small PR-AUC because dominated by precision and with 0.2% positive cases the AUC collapses toward the prevalence.



This means the model's ranking is **well-calibrated** as predicted high risk buckets contain more crashes

Hurdle Model

Brief Description of the Algorithm

Stage 1 — Crash Occurrence (LightGBM Classifier)

- Objective: **binary**
- Key params: LR **0.05**, 31 leaves, feature/bagging frac **0.8**
- Max **2000** trees + early stopping (100 rounds)
- Output: **P(crash > 0)**

Stage 2 — Crash Count Given Occurrence (LightGBM Poisson Regressor)

- Objective: **poisson**
- Key params: LR **0.05**, 63 leaves, feature/bagging frac **0.8**
- Max **2000** trees + early stopping (100 rounds)
- Trained on **crash > 0** rows
- Output: **E[count | crash > 0]**

Final Prediction

$$\lambda = P(crash > 0) \times E[count \mid crash > 0]$$

Model Specifications

Stage One Objective

Binary (for the crash occurrence model)

Stage Two Objective

Poisson (for the conditional crash count model)

Learning Rate

0.05 (used in both models to control the step size at each iteration)

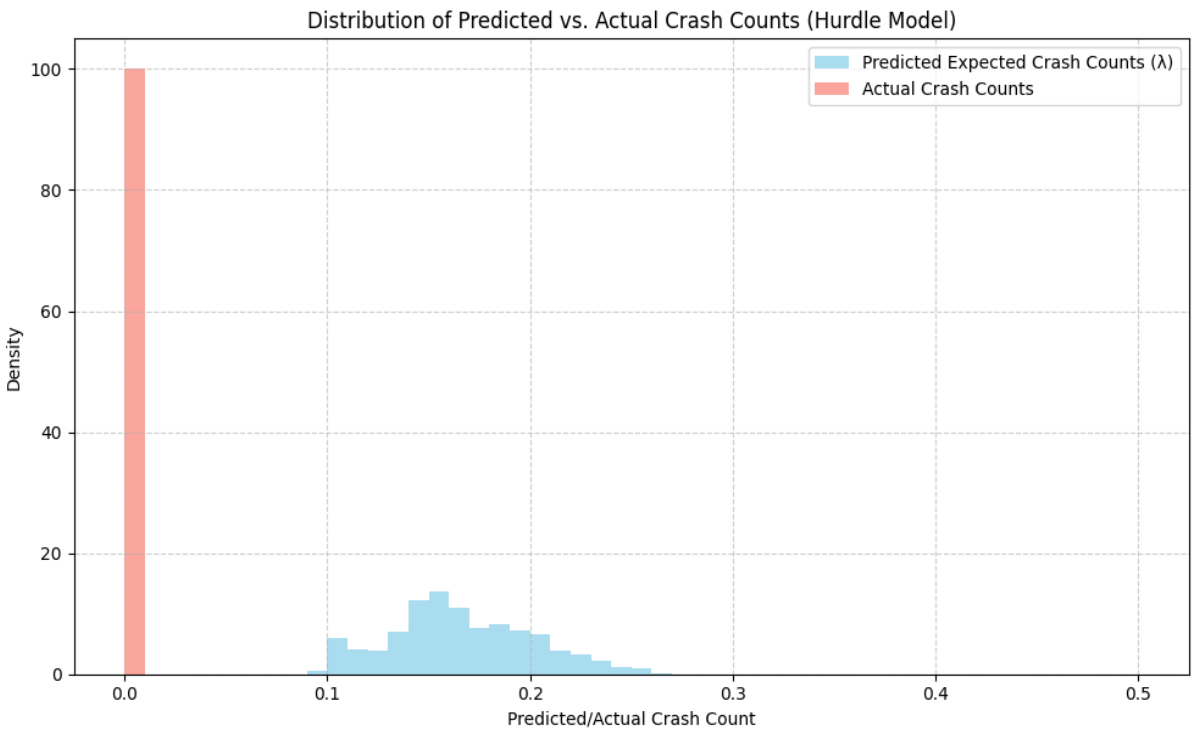
Number of Estimators with Early Stopping

2000 with stopping_Rounds=100

Tree Complexity (num_leaves)

This parameter controls the maximum number of leaves in one tree. It was set to 31 for the binary model and 63 for the count model

Key Insights



1. Actual crash counts (red) are almost always zero - **crashes are extremely rare**

2. The hurdle model's predicted counts (blue) are much higher because it always **predicts "1 crash given a crash,"** and Stage 1 probabilities aren't calibrated to this ultra-rare base rate.

3. The mismatch is expected - there simply **isn't enough variation** in crash counts for a meaningful count model to learn from.

The real predictive insight comes from the crash-occurrence model which ranks **when and where crashes are likely**

Key Takeaways & Next Steps

Takeaways

- Austin's 'dangerous' locations are not only defined by high crash counts. Their risks spike when the current conditions resemble those that historically led to crashes
- High speed corridors like I-35 remain high-risk even in "safe" moments, suggesting that infrastructure plays a major role beyond driver behavior.
- Crash Risk in Austin is defined by transitions, not static states. Risk jumps most sharply during changing conditions (start of rush hour, weather).
- Austin's nightlife corridors become safer after midnight. The crash-occurance model shows a drop in risk despite people assuming late hours are more dangerous.

Next Steps

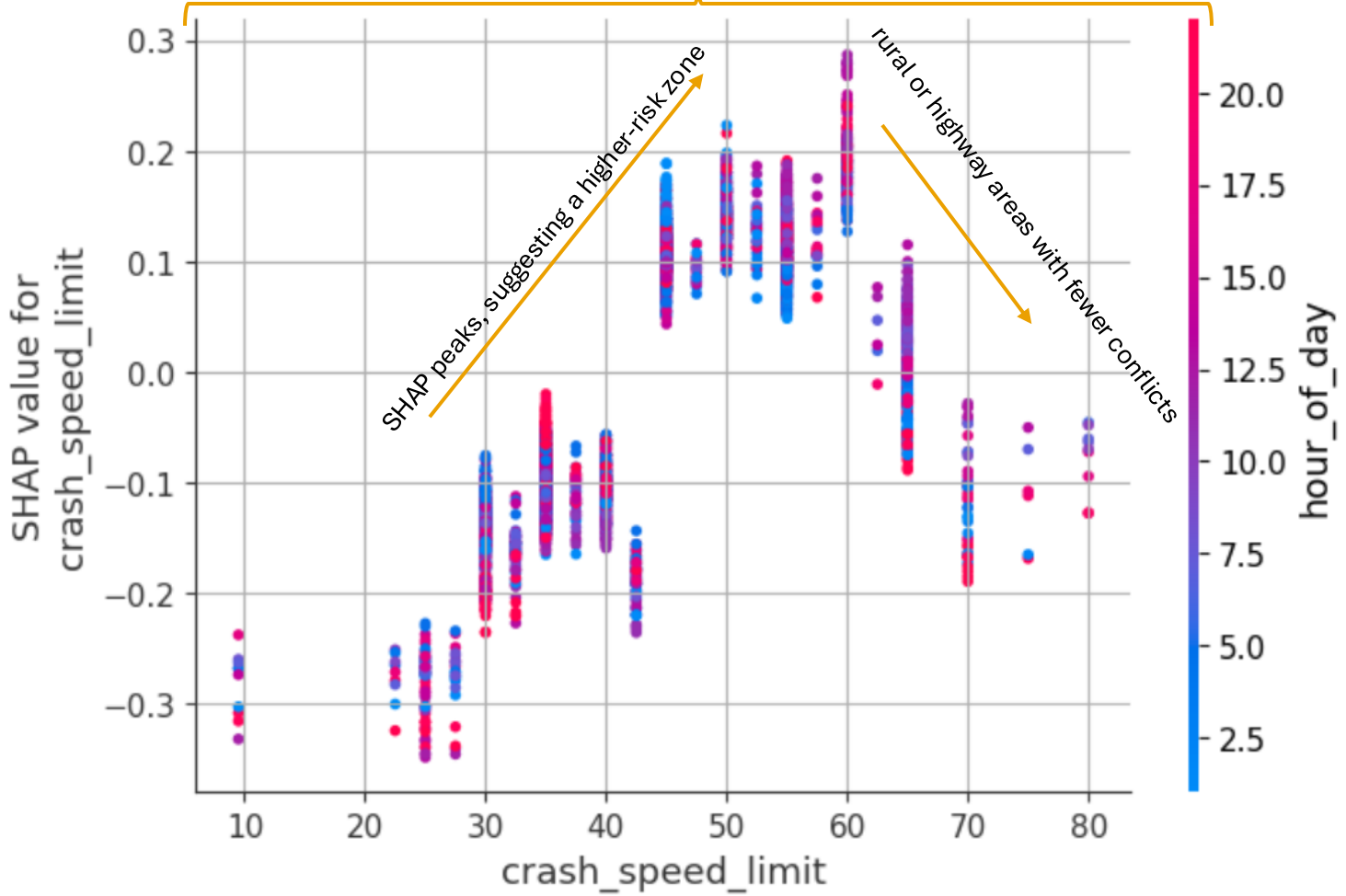
- Integrate weather and major event data to capture short-term risk spikes
- Develop an interactive risk map for easier exploration and policy use
- Incorporate real-time traffic data for dynamic, updated risk predictions

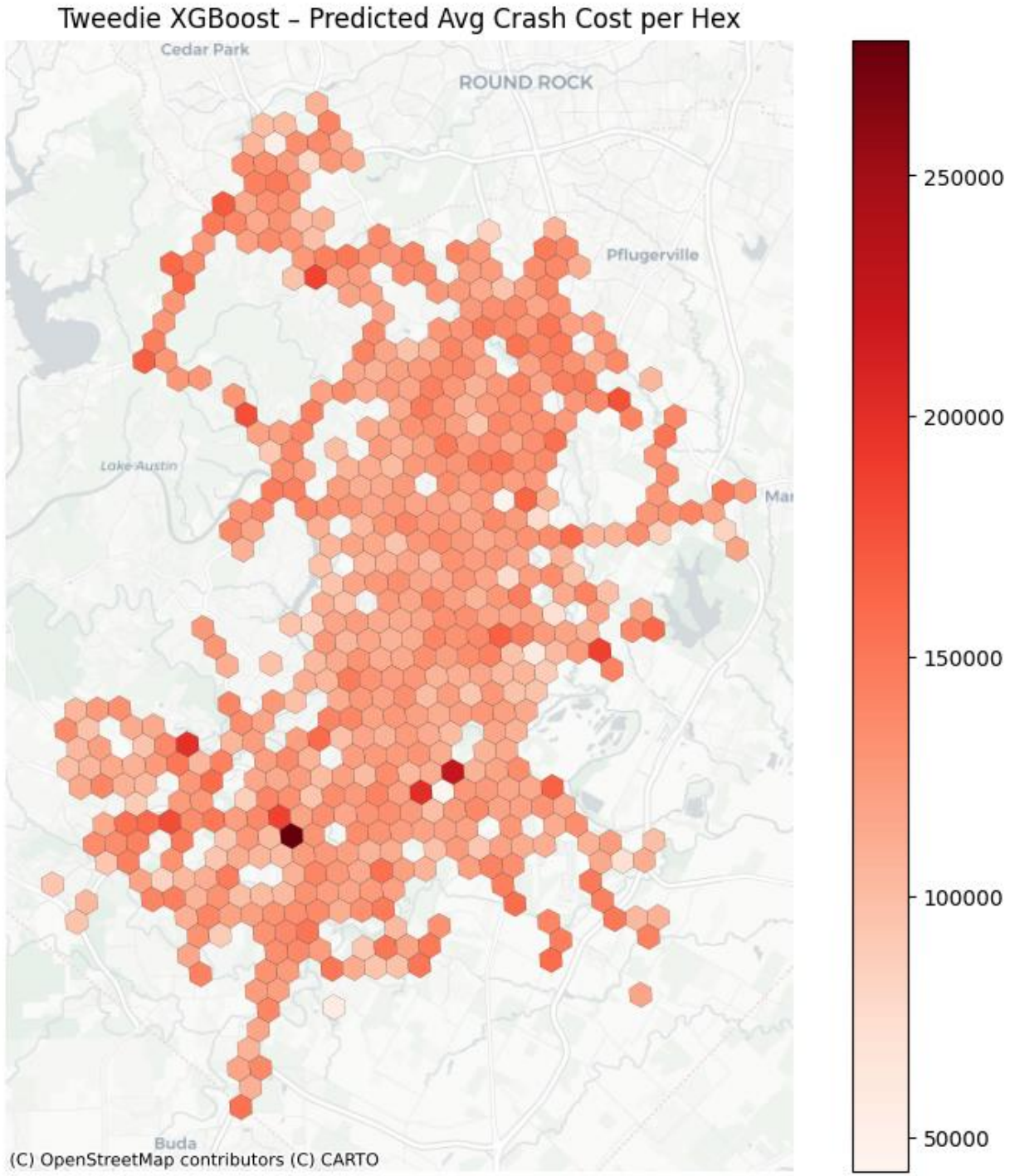
Questions?

Appendix

At higher speed limits, the model tends to assign higher crash risk, especially during busy daytime hours

At medium/high speed limits, **red dots (later hours)** show higher crash risk





Crash Cost Hotspot Map

- This map shows the model's **estimated relative crash severity patterns** across Austin
- Darker hexagons indicate areas where conditions historically align with **higher-cost crashes**
- High-severity regions overlap with major corridors such as **I-35, US-183, and central Austin arterials**
- Spatial patterns reinforce that severity risk is influenced by **speed environment, roadway design, and traffic-flow context**
- This visualization supports the **exploratory purpose** of the Tweedie XGBoost model—providing **severity tendencies**, not precise cost predictions